RVCalPy radial-velocity benchmark report

Broadening function: LL Aqr and AK For · Cross-correlation: 16 UVES high-velocity stars
RVCalPy, https://github.com/atemelci/RVCalPy · 2026 October 24

We compare RVCalPy velocities with published velocities of the same observations (LL Aqr, AK For) and of the same spectra (UVES sample). With the broadening function, the LL Aqr residuals scatter by 0.092 km s^{-1} for the primary and 0.142 km s^{-1} for the secondary (χ^2_r = 1.03 and 0.81). The AK For residuals scatter by 0.150 and 0.864 km s^{-1} (χ^2_r = 0.45 and 2.54), with slopes of 0.9986 and 0.9923. With cross-correlation, the 16 UVES velocities differ from the published ones by a median |Δ| of 0.483 km s^{-1} and a weighted offset of +0.172 ± 0.125 km s^{-1}. Over the 919 km s^{-1} velocity span the slope is 1.000170 ± 0.000479.

1. What was tested

	LL Aqr	AK For	UVES sample
Method	Broadening function	Broadening function	Cross-correlation
Reference	Graczyk et al. (2016)	Hełminiak et al. (2014)	Alazzawi et al. (2026)
Instrument	HARPS	FEROS (14) + HARPS (2)	UVES, DIC1 390+580, red arm
Measured / compared	17 / 16 observations	16 / 14 observations	16 / 16 stars, one epoch each
Template	HIP 57757 (6093 K)	HIP 108870 (4676 K)	synthetic, 5215 K, log g 3.55
Window / step	500–550 nm, Δv = 1 km s^{-1}	500–550 nm, Δv = 2 km s^{-1}	500–550 nm
Period / velocity span	P = 20.1784 d	P = 3.98099 d	919 km s^{-1}

Table 1. Material and settings. All spectra are public ESO archive phase-3 products; only epochs present in both data sets are compared.

2. How the numbers are defined

For each matched epoch we take the difference between our velocity and the published one, and add the two uncertainties in quadrature. Δ is the weighted mean offset, s the scatter, and χ^2_r the reduced chi-square about Δ. The slope a and intercept b come from a weighted straight-line fit, and a − 1 is quoted as the amplitude error (BF) or the velocity-scale error (CCF):

$$d_i = v_i^{\mathrm{ours}} - v_i^{\mathrm{pub}}, \quad \sigma_i = \sqrt{\varepsilon_i^{2,\mathrm{ours}} + \varepsilon_i^{2,\mathrm{pub}}}, \quad w_i = \sigma_i^{-2}$$

$$\Delta = \frac{\sum w_i d_i}{\sum w_i}, \quad s = \sqrt{\frac{\sum (d_i - \bar{d})^2}{N-1}}, \quad \chi^2_r = \frac{1}{N-1}\sum \left(\frac{d_i - \Delta}{\sigma_i}\right)^2, \quad v^{\mathrm{ours}} = a\, v^{\mathrm{pub}} + b$$

No χ^2_r is given for the UVES comparison: both velocities are measured from the same exposures and no per-star error is published (the paper quotes a global ~0.5 km s^{-1}).

3. Broadening function: LL Aqr and AK For

	LL Aqr		AK For	
	Primary	**Secondary**	**Primary**	**Secondary**
Observations compared	16	16	14	14
Offset Δ (km s^{-1})	+0.251 ± 0.023	+0.294 ± 0.040	+0.208 ± 0.059	+0.318 ± 0.130
Scatter s (km s^{-1})	0.092	0.142	0.150	0.864
Reduced χ^2	1.03	0.81	0.45	2.54
Slope a	0.9997	0.9989	0.9986	0.9923
Amplitude error	-0.03%	-0.11%	-0.14%	-0.77%
Authors' own fit rms	0.053	0.049	0.082	0.096

Table 2. Broadening-function velocities against the published ones. The last row is the residual rms of the authors' own orbital fits.

LL Aqr (16 observations). For the primary and secondary respectively, the offsets are +0.251 ± 0.023 and +0.294 ± 0.040 km s^{-1}, the scatter is 0.092 and 0.142 km s^{-1}, and χ^2_r is 1.03 and 0.81. The amplitude errors are -0.03% and -0.11%.

AK For (14 observations: 12 FEROS, 2 HARPS). For the primary and secondary respectively, the offsets are +0.208 ± 0.059 and +0.318 ± 0.130 km s^{-1}, the scatter is 0.150 and 0.864 km s^{-1}, and χ^2_r is 0.45 and 2.54. The amplitude errors are -0.14% and -0.77%. The secondary residuals run from -1.64 km s^{-1} at phase 0.14 to +1.27 km s^{-1} at phase 0.90.

4. Cross-correlation: 16 UVES high-velocity stars

Star	RV, this work [km s⁻¹]	RV, previous work [km s⁻¹]	RV, Gaia DR3 [km s⁻¹]	Δ [km s⁻¹]
TYC 7524–00065-1	+351.487 ± 0.020	+351.920	+350.953 ± 0.217	-0.433
TYC 7535–00160-1	+355.061 ± 0.034	+353.523	+355.069 ± 0.253	+1.538
C0213360-505024	+387.973 ± 0.027	+388.600	+388.222 ± 0.611	-0.627
C1012254-203007	+508.080 ± 0.039	+506.755	+508.138 ± 1.068	+1.325
C1032508-244851	+509.635 ± 0.027	+509.490	+515.354 ± 2.470	+0.145
C1126364-293415	+411.112 ± 0.017	+409.870	+409.674 ± 0.926	+1.242
C1250398-030748	+398.196 ± 0.034	+403.020	+396.221 ± 0.979	-4.824
C1302091-323721	+410.533 ± 0.018	+410.000	+409.817 ± 3.027	+0.533
C1304064-331913	+360.415 ± 0.017	+360.720	+357.825 ± 0.994	-0.305
C1330590-230955	+335.503 ± 0.024	+335.350	+334.919 ± 0.526	+0.153
TYC 7274–00734-1	+408.415 ± 0.016	+408.317	+407.955 ± 0.208	+0.098
C1519196-191359	-409.450 ± 0.021	-409.610	-409.694 ± 6.570	+0.160
C1536201-144228	-329.661 ± 0.027	-329.600	-328.647 ± 2.112	-0.061
C1919566-632839	+330.789 ± 0.023	+328.680	+330.399 ± 0.530	+2.109
C2136142-694908	+383.363 ± 0.013	+381.640	+382.882 ± 0.330	+1.723
TYC 8019–00159-1	+99.092 ± 0.034	+99.117	+98.450 ± 0.171	-0.025

Table 3. Rows follow the order of Table 3 in Alazzawi et al. (2026). Our velocities are barycentric, measured with RVCalPy-23 on the cleaned spectrum; ± is the width error of the CCF peak (Zucker 2003), which counts the photon noise only. The 2026 September 30 version quoted the formal error of the Gaussian fit and computed v_{bary} at the start of the exposure; the velocities have moved by -120 to -5 m s⁻¹ since. "RV, previous work" is the template-matching velocity of Alazzawi et al. (2026) from the same spectrum, for which no per-star error is published, and Δ is this work minus that velocity. The Gaia DR3 velocity is listed for reference only and enters no statistic.

	All 16 stars
Weighted offset Δ (km s⁻¹)	+0.172 ± 0.125
Median |Δ| (km s⁻¹)	0.483
Scatter s (km s⁻¹)	1.569
Within 1 km s⁻¹	10 of 16
Velocity range (km s⁻¹)	-409.4 to +509.6
Slope a	1.000170 ± 0.000479
Velocity-scale error a − 1	+0.017 ± 0.048%

Table 4. Summary of Table 3.

All 16 stars. The weighted offset is +0.172 ± 0.125 km s⁻¹, the median |Δ| is 0.483 km s⁻¹, and the scatter is 1.569 km s⁻¹. 8 stars agree within 0.5 km s⁻¹ and 10 within 1 km s⁻¹. Across -409.450 to +509.635 km s⁻¹ (919 km s⁻¹), the slope is 1.000170 ± 0.000479, a velocity-scale error of +0.017 ± 0.048%.

References

Alazzawi, J. M. K., Caffau, E., Bonifacio, P., et al. 2026, A&A, DOI 10.1051/0004-6361/202659789

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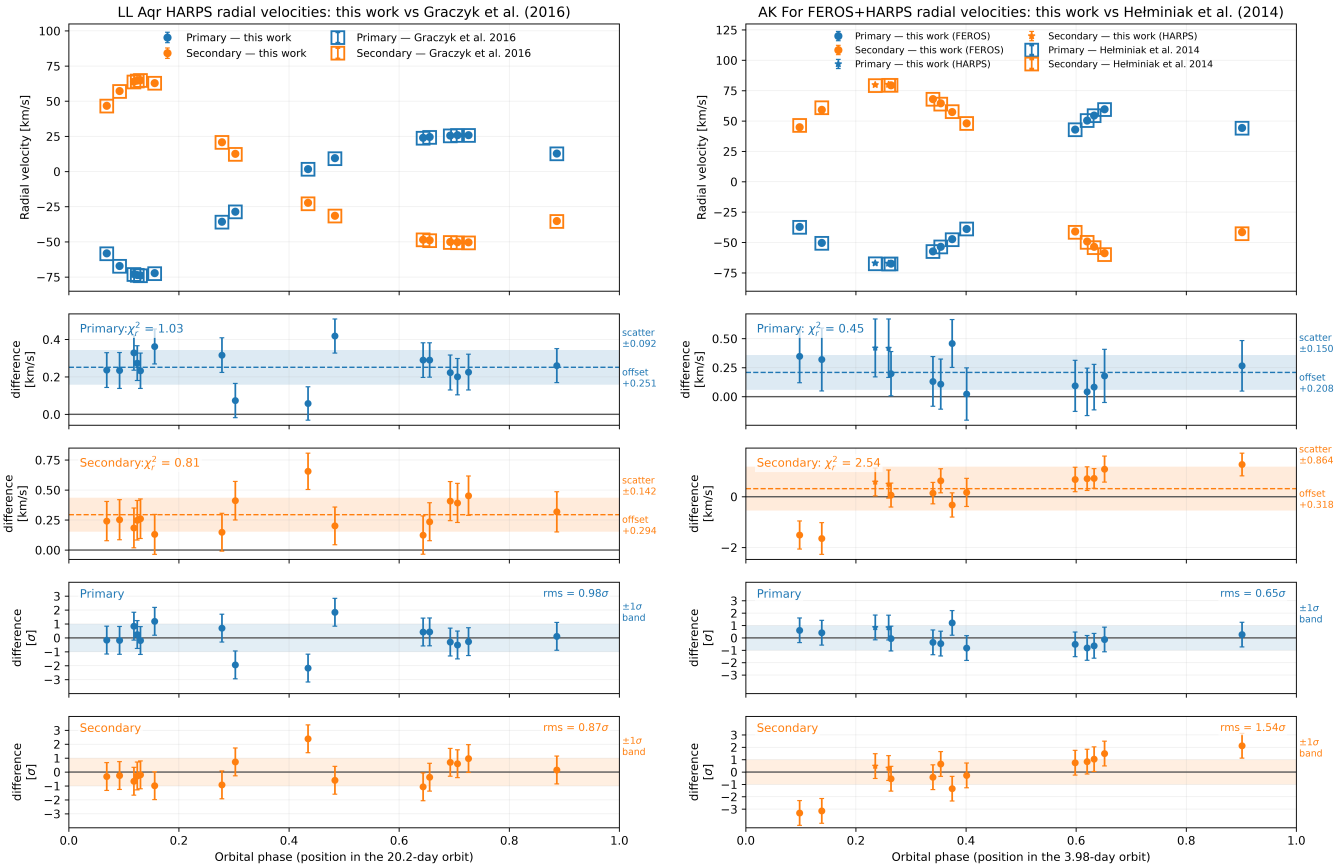


Figure 1. Broadening-function velocities against the published ones: LL Aqr versus Graczyk et al. 2016 (left, 16 observations) and AK For versus Hełminiak et al. 2014 (right, 14 observations; circles FEROS, stars HARPS). Top: phased velocity curves (filled: this work, open: published). Middle: differences with the weighted offset (dashed) and scatter (band). Bottom: residuals in units of their own 1σ after removing the offset.